DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2023

Course: B. Tech. **Branch: Civil Engineering** **Semester: VII**

Subject Code & Name: BTCVC703 Water Resource Engineering

Max Marks: 60 **Date: 18/07/2023** **Duration: 3 Hrs.**

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	**Solve the following questions.**		**12**
A)	Define base period, delta and duty. What are the factors affecting duty?	**CO1**	**06**
B)	Explain classes of soil water with neat labelled sketches	**CO2**	**06**
Q.2	**Solve Any Two of the following.**		**12**
A)	Define Arch dam. Describe various types of arch dam with suitable diagram.	**CO2**	**06**
B)	Explain the various levels in reservoir with suitable diagram.	**CO2**	**06**
C)	Write down the factors affecting selection of dam for available site conditions.	**CO2**	**06**
Q. 3	**Solve Any Two of the following.**		**12**
A)	What do you mean by spillways? Why are spillways provided in dams? Enlist the types of spillways.	**CO2**	**06**
B)	Explain Bligh's creep theory and state its limitations.	**CO2**	**06**
C)	Why should lining be provided in canals? What are the merits and demerits of canal lining?	**CO2**	**06**
Q.4	**Solve Any Two of the following.**		**12**
A)	Explain the types of tube wells with neat sketches.	**CO1**	**06**
B)	Define the following terms: 1. Specific yield 2. Unit Hydrograph 3. Aquifuges 4. Yield of well 5. Catchment area 6. Infiltration	**CO1**	**06**
C)	What are the different types of precipitation? Describe with neat sketches.	**CO 2**	**06**
Q. 5	**Solve the following questions.**		**12**
A)	How to reclaim the water-logged alkaline sands.	**CO 1**	**06**
B)	What are the preventive and curative measures of water logging?	**CO1**	**06**

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